LITERATURE REVIEW

Intelligent Link Integrity Management and Autonomous Remediation Framework for Enterprise Document Repositories

Research papers and literature mapping for the Design Project

Prepared for Design Project Report

1. Overview

This literature review covers the major research areas underlying the proposed project: link rot and link integrity, automatic link recovery, web-resource recommendation, traceability-link recovery, semantic similarity, semantic retrieval, and document-level traceability. The literature indicates that these capabilities have largely been studied as separate problems. The proposed project combines them into an integrated framework for enterprise document repositories.

2. Recommended Research Papers

2.1 Link-Rot in Web-Sourced Multimedia Datasets

Authors: Viktor Lakic, Luca Rossetto, Abraham Bernstein
Year: 2023
DOI: 10.1007/978-3-031-27077-2_37

Methodology: Investigates link rot in datasets containing web-sourced multimedia resources and examines the persistence and availability of external links.

Results: Provides empirical evidence that links in datasets can become unavailable over time, demonstrating that link rot is a continuing data-integrity problem.

Limitations: Primarily studies and detects link rot; it does not provide a complete autonomous repair workflow.

Relation to the proposed project: Supports the problem definition and motivates continuous link-integrity monitoring in the proposed framework.

Paper: https://dblp.org/rec/conf/mmm/LakicRB23

2.2 Archiving website-based references in academic papers: Problems caused by reference rot, potential solutions and limitations

Authors: J. Teixeira da Silva et al.
Year: 2023
DOI: 10.1002/leap.1560

Methodology: Examines reference rot in academic publications and discusses web archiving as a way to preserve inaccessible online references.

Results: Shows that reference rot threatens the long-term integrity and accessibility of web-based references.

Limitations: Archiving preserves historical resources but does not necessarily identify the best current replacement URL.

Relation to the proposed project: Supports the need to move beyond preservation toward replacement discovery and remediation.

Paper: https://onlinelibrary.wiley.com/doi/10.1002/leap.1560

2.3 Making Recommendations from Web Archives for 'Lost' Web Pages

Authors: Lulwah M. Alkwai, Michael L. Nelson, Michele C. Weigle
Year: 2020
DOI: 10.48550/arXiv.1908.02819

Methodology: Classifies unavailable URIs, generates candidates, filters archived candidates, and ranks them using URI similarity, temporal similarity, popularity, and archival quality.

Results: For first-level URI classification, the reported approach achieved an F1 score of 0.59 using all-grams after removing numerals and top-level domains.

Limitations: Focused on web archives and archived resources rather than heterogeneous enterprise PDF/DOCX repositories.

Relation to the proposed project: Provides a strong conceptual basis for candidate generation and ranking for broken-link replacement.

Paper: https://arxiv.org/abs/1908.02819

2.4 A Literature Review of Automatic Traceability Links Recovery for Software Change Impact Analysis

Authors: Thazin Win Win Aung, Huan Huo, Yulei Sui
Year: 2020
DOI: 10.1145/3387904.3389251

Methodology: Systematic review of automatic traceability-link recovery research, including artifact types, approaches, evaluation methods, and machine/deep-learning techniques.

Results: Reviews 33 relevant studies and identifies automated traceability recovery as an active research area with growing use of machine learning.

Limitations: The review focuses on software-engineering traceability rather than enterprise document hyperlinks.

Relation to the proposed project: Provides the literature foundation for treating broken links as recoverable relationships rather than simple errors.

Paper: https://ieeexplore.ieee.org/document/10186860/

2.5 Automatic Traceability Link Recovery via Active Learning

Authors: Tian-bao Du, Guo-hua Shen, Zhi-qiu Huang, Yao-shen Yu, De-xiang Wu
Year: 2020
DOI: 10.1631/FITEE.1900222

Methodology: Uses active learning to reduce the amount of manually labeled data required for traceability-link recovery and evaluates the approach on seven traceability datasets.

Results: The reported active-learning method achieved higher F-scores than the compared information-retrieval and machine-learning approaches.

Limitations: Designed for software traceability datasets and requires a suitable learning setup.

Relation to the proposed project: Supports a human-in-the-loop design in which approved repairs can become future training data.

Paper: https://jzus.zju.edu.cn/article.php?doi=10.1631/FITEE.1900222

2.6 Traceability Link Recovery between Requirements and Models using an Evolutionary Algorithm Guided by a Learning to Rank Algorithm

Authors: Ana Cristina Marcén, Raúl Lapeña et al.
Year: 2020
DOI: 10.1016/j.jss.2020.110519

Methodology: Combines evolutionary algorithms with learning-to-rank to rank candidate model fragments for requirements.

Results: The reported railway case study achieved mean precision of 59.91%, recall of 78.95%, F-measure of 62.50%, and MCC of 0.64.

Limitations: Addresses requirements-to-model traceability rather than broken web or document hyperlinks.

Relation to the proposed project: Supports confidence-based ranking of possible replacement resources.

Paper: https://www.sciencedirect.com/science/article/pii/S0164121220300029

2.7 Combining Machine Learning and Logical Reasoning to Improve Requirements Traceability Recovery

Authors: Tong Li, Shiheng Wang, David Lillis, Zhen Yang
Year: 2020
DOI: 10.3390/app10207253

Methodology: Combines machine learning with logical reasoning and artifact properties instead of relying only on textual similarity.

Results: Reports improvements from incorporating artifact properties and contextual reasoning into traceability recovery.

Limitations: Evaluated in requirements traceability and not in enterprise document link remediation.

Relation to the proposed project: Supports combining URL, anchor text, surrounding context, metadata, and semantic similarity when scoring replacement candidates.

Paper: https://www.mdpi.com/2076-3417/10/20/7253

2.8 Analyzing Close Relations Between Target Artifacts for Improving IR-Based Requirement Traceability Recovery

Authors: Haijuan Wang, Guohua Shen, Zhiqiu Huang, Yaoshen Yu, Kai Chen
Year: 2021
DOI: 10.1631/FITEE.2000126

Methodology: Uses relationships between target artifacts to improve information-retrieval-based traceability recovery.

Results: Reports precision improvements of 40%, 8%, 20%, 4%, and 6% across five public datasets, with an average improvement of 15.6% under the same recall conditions.

Limitations: Focused on requirements traceability and public software-engineering datasets.

Relation to the proposed project: Supports using the context around a hyperlink, such as section heading, anchor text, paragraph, and document title.

Paper: https://academic.hep.com.cn/fitee/EN/10.1631/FITEE.2000126

2.9 Enhancing Traceability Link Recovery with Unlabeled Data

Authors: Jianfei Zhu, Guanping Xiao, Zheng Zheng, Yulei Sui
Year: 2022
DOI: 10.1109/ISSRE55969.2022.00050

Methodology: Investigates the use of unlabeled data to improve traceability-link recovery and reduce dependence on manually labeled examples.

Results: Addresses the data-labeling challenge in deep-learning-based traceability recovery by exploiting additional unlabeled information.

Limitations: The problem domain is software traceability rather than document hyperlink repair.

Relation to the proposed project: Suggests a future learning loop using historical automatic repairs and human approvals/rejections.

Paper: https://dblp.uni-trier.de/rec/conf/issre/ZhuX0S22.html

2.10 Enhancing Traceability Link Recovery with Fine-Grained Query Expansion Analysis

Authors: Research study on query expansion for traceability recovery
Year: 2023
DOI: 10.3390/info14050270

Methodology: Studies fine-grained query expansion to improve retrieval of relevant traceability links.

Results: Shows that expanding queries with additional relevant terms can improve recovery of related artifacts.

Limitations: Evaluated in traceability recovery rather than enterprise link repair.

Relation to the proposed project: Supports generating richer search queries from anchor text, surrounding text, document title, and URL components when searching for replacements.

Paper: https://www.mdpi.com/2078-2489/14/5/270

2.11 Recovering Trace Links Between Software Documentation and Code

Authors: Jan Keim, Sophie Corallo, Dominik Fuchß, Tobias Hey, Tobias Telge, Anne Koziolek
Year: 2024
DOI: 10.1145/3597503.3639130

Methodology: Introduces TransArC, using software architecture models as intermediate artifacts to bridge documentation and code.

Results: Across five open-source projects, the reported average F1 was 0.98 for model-to-code traceability and 0.82 for documentation-to-code traceability.

Limitations: Requires architecture-model context and targets software documentation/source-code relationships.

Relation to the proposed project: Demonstrates that document context can be used to recover meaningful relationships with external artifacts.

Paper: https://ardoco.de/c/icse24

2.12 Dense Passage Retrieval for Open-Domain Question Answering

Authors: Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, Wen-tau Yih
Year: 2020
DOI: 10.18653/v1/2020.emnlp-main.550

Methodology: Represents queries and passages as dense vectors and retrieves semantically relevant passages using vector similarity.

Results: Reported 9–19 percentage-point improvements over a strong BM25 baseline in top-20 retrieval accuracy across evaluated datasets.

Limitations: Designed for open-domain question answering rather than broken-link remediation.

Relation to the proposed project: Provides a technical foundation for semantic candidate retrieval in the proposed replacement-discovery module.

Paper: https://aclanthology.org/2020.emnlp-main.550/

3. Additional Foundational Paper

Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks

Authors: Nils Reimers, Iryna Gurevych
Year: 2019
DOI: 10.18653/v1/D19-1410

Methodology: Sentence-BERT creates efficient sentence embeddings that can be compared using similarity measures such as cosine similarity.

Results: The work demonstrated that sentence embeddings can make semantic similarity and search substantially more efficient than repeatedly comparing full BERT representations.

Limitations: This is a foundational semantic-embedding method rather than a broken-link repair system.

Relation: It can provide the embedding layer for comparing the context of a broken hyperlink with candidate replacement pages or documents.

Paper: https://aclanthology.org/D19-1410/

4. Mapping of Literature to Proposed System

Proposed module	Supporting literature
Link integrity detection	Lakic et al. (2023); Teixeira da Silva et al. (2023)
Broken URL recovery	Alkwai et al. (2020)
Automatic link/traceability recovery	Aung et al. (2020); Du et al. (2020)
Candidate ranking	Marcén et al. (2020)
Context-aware matching	Li et al. (2020); Wang et al. (2021)
Human-in-the-loop learning	Du et al. (2020)
Use of unlabeled historical data	Zhu et al. (2022)
Query expansion	2023 fine-grained query-expansion study
Document semantic relationships	Keim et al. (2024)
Semantic retrieval	Karpukhin et al. (2020)
Semantic similarity	Reimers & Gurevych (2019)

5. Research Gap

Existing research has extensively investigated the detection of link rot, recommendation of alternative web resources, semantic retrieval, and automated traceability-link recovery. However, these capabilities are generally studied as separate problems and are commonly evaluated on webpages, web archives, or software-engineering artifacts. There is comparatively less work on an integrated framework specifically designed for enterprise document repositories that can extract hyperlinks from heterogeneous documents, detect link failures, use document context and semantic similarity to identify replacement resources, assign confidence to proposed replacements, and autonomously remediate high-confidence links while maintaining an audit trail and human-review mechanism.

6. Suggested Literature Review Structure

• 2.1 Link Rot and Link Integrity: discuss link degradation, reference rot, and continuous integrity monitoring.

• 2.2 Automated Link and Traceability Recovery: discuss automatic recovery, active learning, learning-to-rank, and traceability research.

• 2.3 Contextual and Semantic Retrieval: discuss semantic similarity, dense retrieval, query expansion, and contextual matching.

• 2.4 Document-Level Traceability: discuss relationships between documentation and other artifacts.

• 2.5 Research Gap: explain how the proposed project integrates these capabilities into one enterprise document remediation workflow.

7. Reference List — Clickable Links

Lakic, V., Rossetto, L., & Bernstein, A. (2023). Link-Rot in Web-Sourced Multimedia Datasets.
https://dblp.org/rec/conf/mmm/LakicRB23

Teixeira da Silva, J. et al. (2023). Archiving website-based references in academic papers.
https://onlinelibrary.wiley.com/doi/10.1002/leap.1560

Alkwai, L. M., Nelson, M. L., & Weigle, M. C. (2020). Making Recommendations from Web Archives for 'Lost' Web Pages.
https://arxiv.org/abs/1908.02819

Aung, T. W. W., Huo, H., & Sui, Y. (2020). A Literature Review of Automatic Traceability Links Recovery.
https://ieeexplore.ieee.org/document/10186860/

Du, T.-b. et al. (2020). Automatic Traceability Link Recovery via Active Learning.
https://jzus.zju.edu.cn/article.php?doi=10.1631/FITEE.1900222

Marcén, A. C., Lapeña, R. et al. (2020). Traceability Link Recovery between Requirements and Models.
https://www.sciencedirect.com/science/article/pii/S0164121220300029

Li, T., Wang, S., Lillis, D., & Yang, Z. (2020). Combining Machine Learning and Logical Reasoning.
https://www.mdpi.com/2076-3417/10/20/7253

Wang, H. et al. (2021). Analyzing Close Relations Between Target Artifacts.
https://academic.hep.com.cn/fitee/EN/10.1631/FITEE.2000126

Zhu, J. et al. (2022). Enhancing Traceability Link Recovery with Unlabeled Data.
https://dblp.uni-trier.de/rec/conf/issre/ZhuX0S22.html

Enhancing Traceability Link Recovery with Fine-Grained Query Expansion Analysis. (2023).
https://www.mdpi.com/2078-2489/14/5/270

Keim, J. et al. (2024). Recovering Trace Links Between Software Documentation and Code.
https://ardoco.de/c/icse24

Karpukhin, V. et al. (2020). Dense Passage Retrieval for Open-Domain Question Answering.
https://aclanthology.org/2020.emnlp-main.550/

Reimers, N. & Gurevych, I. (2019). Sentence-BERT.
https://aclanthology.org/D19-1410/

Note: Verify final bibliographic metadata against the publisher/DOI record before submission.